Toolshed User Manual

Service facilities, fuels, interchanges, and couplers for Railroader

Contents

Getting Started

Requirements

- Railroader `2025.1.x`
- Unity Mod Manager
- FUSE — Toolshed requires it and loads after it

Install

1. Install Unity Mod Manager for Railroader.
2. Install FUSE and confirm it loads.
3. Place the `Toolshed` folder in `Railroader/Mods`.
4. Start Railroader and load a map.

Toolshed declares its own asset packs (`SCAssetPacks`) in its manifest, so FUSE mounts the models automatically. There is nothing separate to install.

Verify

Open the in-game console and run:

```
/fuse.loaded
```

Both `FUSE` and `Toolshed` should be listed and applied. If Toolshed is missing, it was never discovered — check that the folder is in `Railroader/Mods` and enabled in Unity Mod Manager. If it is listed as faulted, check `FUSE.log` for the phase that failed.

Because Toolshed's content is delivered through FUSE, `/fuse.assets` will also show its asset pack folders once they are mounted.

What You Get

Toolshed does not change anything on its own — it adds capabilities that route packages and locomotives use.

- **Service facilities** appear where a route package places them. On a route without any, you will not see coal towers or water columns simply from installing Toolshed. See Service Facilities.
- **Oil and wood firing** applies to locomotives whose definitions declare those fuels. See Oil & Wood Firing.
- **Selective interchanges** are configured by route packages. See Selective Interchanges.
- **Link and pin couplers** apply to equipment that uses them, and are tunable in the Unity Mod Manager settings panel. See Link & Pin.
- **Hand turntables** are placed by route packages.

If you installed Toolshed expecting visible changes on a stock map, this is why nothing looks different: it is a toolkit for route content, not a standalone content pack.

Settings

Toolshed's settings are in the Unity Mod Manager panel. The link-and-pin coupler spacing values live there:

Setting	Default
Equipment separation	`0.84`
Caboose separation	`0.84`
Locomotive/tender separation	`0.93`
Slack	`0.02`

See Link & Pin Couplers for what these do.

Where The Files Are

What	Where
Log	`%USERPROFILE%\AppData\LocalLow\Giraffe Lab LLC\Railroader\FUSE.log`
Unity log	`%USERPROFILE%\AppData\LocalLow\Giraffe Lab LLC\Railroader\Player.log`
Mod folder	`Railroader\Mods\Toolshed`

Reporting Problems

Include `FUSE.log`, `Player.log`, the output of `/fuse.loaded`, your mod list, and the route package involved. Toolshed problems are usually route-content problems, so naming the route matters.

Issues: <https://github.com/Hrogers-Rog/TheToolShed/issues>

Service Facilities

Working fuel and water stops. A Toolshed service facility is a placed structure — coal tower, water column, diesel or bunker-C stand, wood shed — that actually transfers its load into a spotted locomotive or tender, with animation and particle effects while it runs.

Supported Loads

Load	Id
Coal	`coal`
Water	`water`
Diesel fuel	`diesel-fuel`
Bunker C	`bunker-c`
Wood	`pulpwood`

Wood uses the `pulpwood` load id rather than a Toolshed-specific one, so it interoperates with existing pulpwood industries and cars.

Using One

1. Spot the locomotive or tender so the fuel point sits on the facility's load point span.
2. The facility begins transferring on its own.
3. Watch the animation and particles — they run while the transfer is active and stop when it completes.

Two things must be true for anything to happen: the car has to actually be on the load point's span, and the facility has to have fuel available. A facility backed by finite storage stops when its storage is empty; an infinite one never does.

Liquid loads default to a transfer rate of 60 units per unit time.

Storage

A facility is either **infinite** or **finite**.

Infinite facilities — typically water — always have supply. Finite ones draw down a storage component that has to be refilled, usually by delivering the fuel in cars. A coal tower that has run out needs a coal delivery before it will fuel anything again.

Finite storage is what makes fuel logistics part of the operation rather than a formality. It is also the most common reason a facility "stops working."

Troubleshooting

Nothing happens when I stop at the facility. Check the spot first. The fuel point on the car has to be within the load point's span, which is narrower than the visible structure. Then check whether the facility has storage remaining.

It worked before and stopped. Finite storage is empty. Deliver more of that load.

The animation runs but nothing transfers. The car may not accept that load — a coal tender will not take bunker C. Confirm the load id matches what the equipment carries.

The structure is missing entirely. That is an asset or route problem rather than a Toolshed one. Run `/fuse.assets` to confirm the pack mounted, and `/fuse.loaded` to confirm the route package applied.

Authoring

Placing service facilities in your own route package is documented in depth in the reference guides:

- `../Examples/service-facility-setup-guide.md` — the full setup flow: authoring paths, editor component setup, runtime and prefab components, the prefab contract, the FUSE pattern, finite storage, interchange purchase setup, and a release checklist.
- `../Examples/service-loader-component-blueprint.md` — every `ToolshedServiceStorage` and `ToolshedServiceLoadPoint` setting in component order, with recommended values for coal, water, diesel, bunker-C, and wood.

The blueprint is the one to keep open while configuring: it gives concrete recommended values per fuel type rather than describing the fields abstractly.

Common configurations covered there include infinite water, finite bunker-C, diesel fuel, coal, and wood, plus how to repurpose an existing stand loader (for example, a diesel stand into a bunker-C stand).

Related

- Getting Started
- Oil & Wood Firing — the locomotives that consume these fuels

Oil and Wood Firing

Steam locomotives in Railroader burn coal. Toolshed adds oil (bunker C) and wood firing, each with its own consumption model, fuel gauges, and alerter behavior.

Fuels

Fuel	Load id	Notes
Bunker C	`bunker-c`	Heavy fuel oil, measured in gallons
Wood	`pulpwood`	Uses the existing pulpwood load, so it works with pulpwood industries and cars
Coal	`coal`	Base game, unchanged

Wood deliberately reuses `pulpwood` rather than inventing a Toolshed-only load. A wood-burner can be fueled from the same pulpwood your route already hauls.

Conversion Rates

Toolshed converts between fuels so a locomotive's existing coal-based capacity and consumption still mean something:

- **Bunker C**: `0.043` gallons per pound of coal displaced. Density is 62 lb per cubic foot, which works out to roughly 8.29 lb per gallon.
- **Wood**: `1.8` pounds of wood per pound of coal — wood carries less energy per pound, so a wood-burner goes through noticeably more of it.

That 1.8 ratio is the number to keep in mind when planning a wood-fired operation: tender capacity that lasted a full run on coal will not.

Bunker C loads at 60 gallons per second at a service stand.

Running One

From the cab, an oil- or wood-fired locomotive behaves like a coal burner — the fireman's job is the same and the fuel gauge reads the same way. What changes is where you refuel and how fast the supply drops.

The auto-engineer's fuel alerter accounts for the active fuel, so low-fuel warnings fire against the real remaining supply rather than a coal-equivalent figure.

Fueling

Oil and wood locomotives refuel at service facilities that carry the matching load — a bunker-C stand or a wood shed. A coal tower will not fuel them.

If a locomotive sits at a facility and nothing transfers, the usual cause is a fuel mismatch: confirm the facility's load id matches the locomotive's fuel.

Which Locomotives

Toolshed does not convert locomotives on its own. A locomotive burns oil or wood only when its **definition declares that fuel**, which is the job of the equipment mod or route package that ships it.

Installing Toolshed will not change any base-game locomotive. If you expected a locomotive to burn oil and it still burns coal, that locomotive's definition does not declare the fuel.

Toolshed resolves the fuel from the locomotive, and falls back to its fuel car (tender) when the locomotive itself does not declare one — so a tender can carry the fuel definition.

Troubleshooting

My locomotive still burns coal. Its definition does not declare oil or wood. That is an equipment-mod question, not a Toolshed one.

It won't take fuel at a stand. The stand's load has to match the locomotive's fuel. Bunker C and diesel fuel are different loads.

Wood runs out much faster than coal did. Expected — see the 1.8:1 ratio above.

The fuel gauge looks wrong. Check `FUSE.log`. Toolshed logs fuel-resolution problems once per piece of equipment rather than every frame, so search for the locomotive's name.

Related

- Service Facilities — where these fuels come from
- Getting Started

Link and Pin Couplers

Period link-and-pin coupler visuals for locomotives, tenders, and cars, with configurable spacing and slack.

Toolshed's link-and-pin support is **definition-driven**. It changes an end of a vehicle only when that vehicle has a `DetailModel` component using the `LinkAndPinCoupler/link-pin-coupler` asset. It does not replace every coupler in the game.

What It Does

On a marked end, Toolshed:

- Hides the vanilla knuckle coupler on that end only
- Shows the link-and-pin asset as a loose link
- Hides the imported coupler pocket and pin from the asset, so the model can use its own pocket or drawbar
- Leaves only one loose link visible when two link-and-pin ends couple together
- Adds a small slack adjustment to marked ends
- Adds customization checkboxes for every marked end

Player Customization

The equipment customization window exposes the parts per end. With one marked end:

- `Loose Link`
- `Pin`
- `Pocket`

With both ends marked, each is prefixed `A-End` or `B-End`.

The loose link defaults **on**, so existing setups keep working. The imported pin and pocket default **off**, because many locomotive and tender models already include their own pin or pocket detail — turning them on usually double-renders the part. These choices save per car.

Spacing And Slack

Four values in the Unity Mod Manager settings panel control how close link-and-pin equipment couples:

Setting	Default	Applies to
Equipment separation	`0.84`	General equipment
Caboose separation	`0.84`	Cabooses
Locomotive/tender separation	`0.93`	Locomotive and tender pairs
Slack	`0.02`	Free play in the coupling

Locomotive/tender spacing is wider by default because that pairing usually has a drawbar rather than a link.

These are visual and slack-behavior values. Raise separation if your models intersect when coupled; lower it if the gap looks too wide.

Authoring

Full setup is in `../Examples/link-pin-coupler-readme.md`. The short version:

End Naming

The component name tells Toolshed which end it belongs to:

- `Link And Pin Coupler F` — front / A end
- `Link And Pin Coupler R` — rear / B end

Names ending in `Front` or `Rear` also work.

Definition Component

Paste an object from `../Examples/link-pin-coupler-detail-model.json` into the vehicle's `components` array, then tune `parent.path`, `transform.position`, `transform.rotation`, and `transform.scale`.

For a locomotive with a fixed front drawbar, put the visual link only on the tender or rear pocket end. For freight cars, add one component per end that should have link-and-pin visuals.

Asset Ids Must Match

```
"model": {
  "assetPackIdentifier": "LinkAndPinCoupler",
  "assetIdentifier": "link-pin-coupler"
}
```

If those ids change, Toolshed treats the component as an ordinary detail model and will **not** hide the vanilla coupler for that end. This is the most common reason a link-and-pin setup renders both couplers at once.

Related

- Getting Started — where the settings panel lives

Selective Interchanges

A vanilla Railroader interchange is generic — cars appear and disappear. A Toolshed selective interchange turns that same track into a specific off-map industry or connecting railroad, with traffic that makes sense for what it represents.

The worked example is the Ocona-Lufty RR at Smokemont behaving like an off-map sawmill: empty flats and boxcars are ordered in, occasional loaded building-supplies cars are delivered to the mill, the cars go off map, and loaded lumber and poles return at the same interchange to be waybilled onward.

For Players

You do not configure these — route packages do. What you notice is that an interchange starts asking for particular cars and returning particular loads, instead of accepting anything.

If an interchange is behaving oddly, the route package that configured it is where the problem lives. Include the route name in any report.

Authoring

Config Location

Put a config file in either place inside your mod folder:

- `ToolshedSelectiveInterchanges.json` at the mod folder root
- any `.json` file inside a `SelectiveInterchanges` folder

Toolshed scans installed mod folders and applies matching definitions **after** the game has created the industries.

The Traffic Model

A selective interchange has four independent traffic types. They compose — use only the ones your scenario needs.

Field	Purpose
`emptyCarRequests`	Ask for empty cars to send to the off-map operation
`inboundLoads`	Accept loaded supplies the off-map operation consumes
`purchaseLoads`	Create vanilla buy/import loaders for loads the railroad can purchase
`outboundRoutes`	Create loaded-car orders at the interchange for off-map production returning

Two details worth getting right:

`emptyCarRequests` **needs an origin.** For interchange-to-interchange traffic, set `originWeights` or another origin field so the empties spawn at the active external interchange rather than at the off-map interchange itself. Without it the cars appear where they are going, which is not what you want.

`inboundLoads` **only acts when served.** By default Toolshed handles those cars only when the interchange is actually being served, so cars sitting on the interchange track are not removed by the normal industry tick.

`purchaseLoads` **is for buyable goods only.** Use it for what the railroad can buy from the connecting railroad — Ocona-Lufty logs for Ravensford, in the worked example. Do not put normal mill production here unless it genuinely should be purchasable; production that returns loaded belongs in `outboundRoutes`.

Reference

- `../Examples/selective-interchange-readme.md` — the full field reference and a minimal working example
- `../Examples/KingG.Appalachian-Railway.ToolshedSelectiveInterchanges.json` — a complete real configuration

There is also a legacy RailLoader build of this feature in `RailLoaderSelectiveInterchanges/` for pre-FUSE installs.

Related

- Getting Started
- Service Facilities